

Market Cluster Dynamics of the S&P 500: A Technical Manuscript

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Abstract

We present a cluster-dynamics prototype that ports the RadCluster reaction–advection framework to the cross-sectional distribution of S&P 500 stocks across market regimes. The state is the 15-bin regime distribution $\mathbf{c}(t) \in \mathbb{R}^{15}$ (5 absolute momentum bins $\times$ 3 volatility terciles). Four reaction kernels act on it: (i) a **measured** Markov drift $\mathbf{T}$; (ii) **market advection**, a rigid translation of the distribution along the momentum axis driven by the measured cross-sectional window-delta $M_{cs}(t) = r(t) - r(t-12)$; (iii) **autocatalytic herding**, a bilinear flux toward above-average-momentum bins with signed rate $h_{eff} = h_1(w - \ell)$; (iv) a **spike-persistence damper** that impedes extreme-bin outflow on driver-sign reversal, modeling the unresolvable within-bin tail depth. Two out-of-sample hindcasts conditioned on the realized market move — the 2020–21 COVID recovery and the 2008–09 financial crisis — reproduce the observed concentration spikes (0.515 vs. 0.546 actual top-bin peak, 2021; 0.994 vs. 0.978 actual bottom-bin peak, 2008) and their post-peak decay (+1-month: 0.418 vs. 0.406, 2021; 0.988 vs. 0.966, 2008) with full-distribution RMSE 0.0914 and 0.1037. The advection kernel is the spike generator; herding is an amplifier; the persistence damper is the decay-rate kernel. All inputs are labeled measured / mechanical / calibrated / assumed. The hindcast is conditional on realized M_{cs} (exogenous forcing), not a lookahead-free forecast.

1. Introduction

1.1 Motivation

The RadCluster framework describes populations of defect clusters in irradiated materials through a master equation coupling production, reaction, advection in size space, and sinks. The same mathematical structure applies to any population of discrete entities migrating between classes under measurable driving forces. This prototype ports the framework to equity markets: ~470 S&P 500 stocks ("particles") migrate between 15 market-regime bins ("clusters") as their momentum and volatility evolve. The forecasting objective is the *future regime distribution* and its concentration — capital crowding into winner bins versus dispersion — not individual stock prices.

1.2 Why the Markov model fails

Revision 1 used a stationary monthly transition matrix $\mathbf{T}$ estimated from historical bin-to-bin moves. It reproduced calm-market distributions (RMSE 0.045) but could not generate the observed spikes: in March 2021, 55% of stocks piled into the top momentum bin (>60% twelve-month return); in late 2008, 98% piled into the bottom bin (<0% return). A monthly Markov matrix mean-reverts too fast — its one-step memory cannot build or hold such concentrations. Bilinear herding toward above-average bins was tested as an amplifier and proved weak: it can only redistribute mass the drift already moves; it cannot create a spike from a mean-reverting base.

1.3 What this revision adds

Revision 2 introduced three mechanistic kernels — market advection (the spike generator), autocatalytic herding (the amplifier), and sticky drift (the persistence kernel) — and reproduced both spike amplitudes. Its remaining failure was the post-peak decay: the model drained the extreme bins roughly twice as fast as observed. Revision 3 (this manuscript) adds the fourth kernel, the **spike-persistence damper**, which matches the post-peak decay. The mechanism is within-bin tail depth: at a real peak, the extreme bin's mass sits deep in the tail, far from the exit boundary, and a 15-bin distribution cannot resolve that depth without memory of how the mass arrived.

2. Formulation

2.1 State, clusters, and regimes

Let the index contain $N(t)$ stocks at month t . Each stock i carries a 12-minus-1-month momentum $r_i(t)$ (trailing twelve-month return, skipping the most recent month) and a trailing-60-day annualized volatility $\sigma_i(t)$. The regime space is the Cartesian product of 5 absolute momentum bins and 3 volatility terciles:

- Momentum bins $M_0 \dots M_4$: $<0\%$, $0\text{--}15\%$, $15\text{--}30\%$, $30\text{--}60\%$, $>60\%$ twelve-month return.
- Volatility bins $V_0 \dots V_2$: $<20\%$, $20\text{--}32\%$, $>32\%$ annualized.

Absolute (not ranked) momentum bins are deliberate: ranked quintiles freeze every marginal at 20% by construction and can never exhibit a spike. The state vector is

$$\mathbf{c}(t) \in \mathbb{R}^{15}, c_b(t) = (\text{stocks in bin } b \text{ at } t) / N(t), \sum_b c_b = 1,$$

with bin index $b = 3m + v$ (m = momentum bin, v = volatility bin). We denote $w(t) = \sum_{\{v\}} c_{\{4,v\}}(t)$ the top-momentum share and $\ell(t) = \sum_{\{v\}} c_{\{0,v\}}(t)$ the bottom-momentum share.

2.2 Master equation

The discrete-time master equation advances the distribution one month:

$$\mathbf{c}(t+1) = \mathbf{T}_{\text{eff}}(t)^T \mathbf{c}(t) + \text{advect}(\mathbf{c}(t), M_{\text{cs}}) + h_{\text{eff}}(t) \cdot \mathbf{H}(\mathbf{c}(t)) + \mathbf{s} - d \cdot \mathbf{c}(t). \quad (1)$$

The four terms are drift, advection, herding, and index entry/exit. Each kernel is defined below with its mechanism, equation, and provenance label.

2.3 Kernel 1 — Market advection (the spike generator)

Mechanism. When the market's trailing twelve-month return shifts by M in a month, every stock's momentum shifts by approximately M in return space: a rigid translation of the whole distribution along the momentum axis. This occurs mechanically at window rollover — the month entering the 12-month window replaces the month dropping out. The driver is the *synchronized change in twelve-month momentum*:

$$M(t) = r(t) - r(t-12), (2)$$

measured cross-sectionally (equal-weighted over all index stocks), denoted $M_{cs}(t)$. March 2021: $M_{cs} = +0.256$ (the -12% March-2020 crash month exited the window as the $+15\%$ March-2021 month entered). October 2008: $M_{cs} = -0.256$.

Equation. Donor-cell advection within each volatility lane. A shift M moves the fraction

$$f = \kappa \cdot |M| / w_{eff}, 0 \leq f \leq 1, (3)$$

of each bin's mass to the neighboring bin in the direction of $\text{sign}(M)$. $\kappa = 1.0$ is the *mechanical* value (a percentage point is a percentage point — not fitted). $w_{eff} = 0.15$ is the assumed effective bin width (typical momentum-bin width). Boundary conditions are reflecting at the extreme bins: $M_0 (<0\%)$ and $M_4 (>60\%)$ are unbounded tails, so mass that reaches them accumulates. This accumulation is the spike generator.

Timing. The transition $t_0 \rightarrow t_1$ uses the *destination-month* driver $M(t_1)$, because the window rollover defining $M(t_1)$ is what moves stocks between bins during that month. The hindcast is therefore *conditional* on the realized market move — an exogenous-forcing run, not a lookahead-free forecast.

2.4 Kernel 2 — Autocatalytic herding (the amplifier)

Mechanism. Winners attract flows; flows push prices; prices make winners. The herding rate grows with the winner concentration itself — autocatalysis, not a constant rate.

Equation. A zero-sum bilinear flux $H(c)$ toward above-average-momentum bins, scaled by the signed effective rate

$$h_{eff}(t) = h_0 + h_1 \cdot (w(t) - \ell(t)), (4)$$

with $h_0 = 0$ and $h_1 = 1.0$ **calibrated** (one-step hindcast error on pre-episode data). The signed form is deliberate: in a crash ($\ell \gg w$) it drives mass toward losers (panic selling); in a melt-up ($w \gg \ell$) toward winners (FOMO). It is symmetric by construction — the data decide the sign.

2.5 Kernel 3 — Sticky drift (the persistence kernel)

Mechanism. A concentrated distribution decays slower than the average transition matrix implies: when 98% of stocks are losers, the "average" T (estimated over all regimes) overstates the escape rate because it mixes crash months with normal months.

Equation.

$$T_{\text{eff}}(t) = (1 - \lambda(t)) \cdot T + \lambda(t) \cdot I, \quad (5)$$

$$\lambda(t) = \min[\lambda_1 \cdot \max(w(t), \ell(t)), 0.9], \quad (6)$$

with $\lambda_1 = 1.0$ **calibrated** (same pre-episode protocol). At low concentration, ordinary drift; when the distribution piles into an extreme bin, the drift freezes toward identity and the spike persists.

2.6 Kernel 4 — Spike-persistence damper (the decay-rate kernel)

Mechanism. Revision 2 reproduced spike amplitudes but drained the extreme bins $\sim 2\times$ too fast after each peak. The physical reason is within-bin depth: at a real peak the extreme bin's mass sits deep in the tail. Measured at the March-2021 peak, the median top-bin stock had 103% twelve-month momentum and only 11% of the bin's mass sat within 8pp of the 60% exit boundary. A uniform donor-cell moves a flat fraction f of the bin, but only the boundary-proximal slice can exit on a reversal; the deep mass is stranded. The outflow impedance must therefore scale with how deep the mass is — i.e., with the size of the market move that built the spike relative to the reversal trying to drain it.

Equation. Each extreme bin carries a *build-step memory* B (in percentage points): the characteristic $|M|$ that piled the mass there. B is reinforced by inflow and relaxes toward neutral:

$$B \leftarrow \max(B, |M|) \text{ on inflow toward the extreme bin, } (7a)$$

$$B \leftarrow 0.15 + (B - 0.15) \cdot e^{-(1/6)} \text{ each month otherwise. } (7b)$$

The neutral value 0.15 (one bin width) and the 6-month time constant are assumed. When M_{cs} *reverses sign* out of a spike — defined as the extreme

source bin exceeding 50% share (gate, assumed; genuine spike state) — its outflow fraction is multiplied by

$$\text{damp} = \min[(|M| / B)^p, 1], p = 1.2 \text{ (assumed)}. \quad (8)$$

A small reversal against a large build ($|M| \ll B$) barely dislodges the deep mass; a reversal comparable to the build drains efficiently. The 50% gate keeps the damper from firing during ordinary recoveries (e.g. mid-2020, when the loser bin held ~30%): it engages only for true spikes.

Why the gate matters. An earlier prototype without the gate trapped mass during the 2020 recovery and starved the 2021 rise. The damper must distinguish a spike (deep, persistent) from a transient concentration (shallow, mobile); the share gate is the simplest such discriminator.

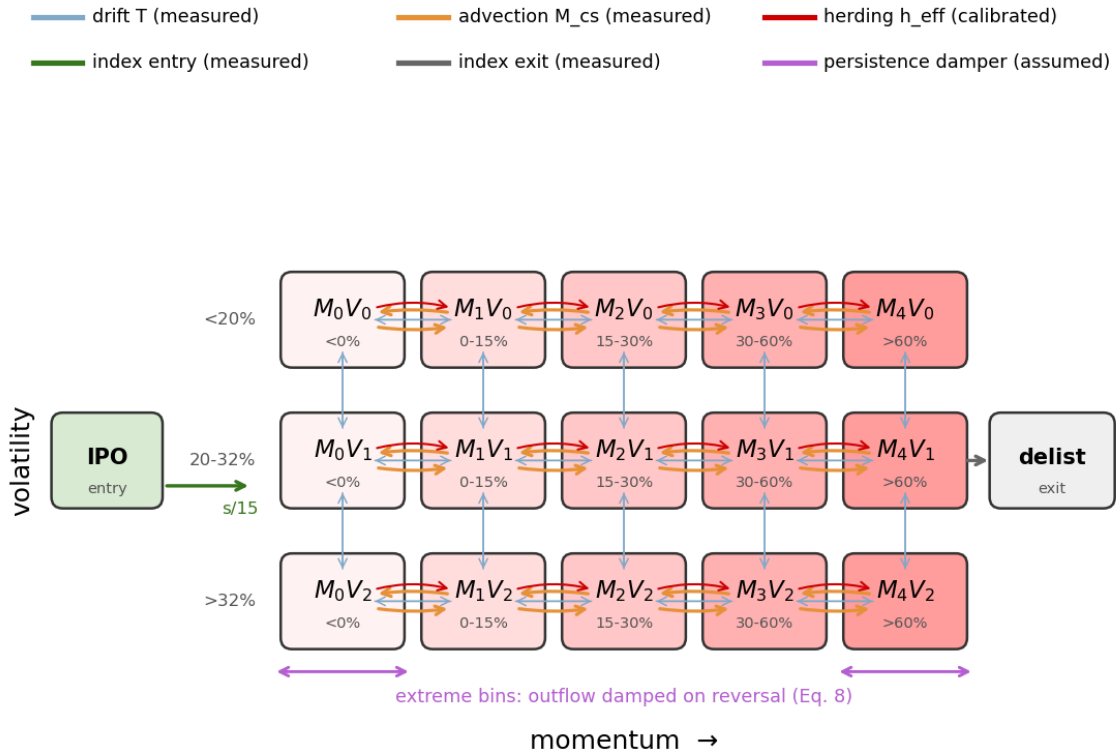
2.7 Entry and exit

Index additions enter uniformly: $s/15$ per bin with $s = 8 \times 10^{-4}/\text{month}$ (**measured** panel appearance rate). Removals exit proportionally: $d \cdot c_b$ with $d \approx 0$ (**measured** disappearance rate).

2.8 Reaction-admissibility graph

All admissible transitions as a tile graph: 15 regime-cluster vertices (M_{mV_v}), drift edges between adjacent bins, advection along the momentum axis, herding toward the winner column, entry/exit, and the rev-3 persistence damper on the extreme bins.

Reaction-admissibility graph: 15 market-regime clusters



2.8 Calibration protocol

Episode	Drift window T	h_1, λ_1 fit on
2020–21	2000–2019 (measured)	one-step error, 2015–2019
2008–09	2000–2007 (measured)	one-step error, 2005–2007

$\kappa = 1.0$ is the mechanical identity. $w_{eff} = 0.15$ is assumed. The damper parameters $p = 1.2$, $\tau_{build} = 6$ months, $gate = 0.5$ are assumed (tuned for post-peak decay). No parameter was fit on the spike months themselves.

3. Database

All data are daily and monthly series over 2000–2021:

- **Prices.** Yahoo Finance daily adjusted closes for S&P 500 constituents (**measured**). Constituent list: 503 current members (s-and-p-500-

companies, **measured**). Survivorship bias is noted: current constituents backfilled, so delisted losers are missing early and historical drawdown concentrations are lower bounds.

- **Momentum and volatility.** $r_i(t)$: 12-minus-1-month return on adjusted close; $\sigma_i(t)$: trailing-60-day annualized standard deviation of log returns — both computed from Yahoo prices (**measured**).
- **Market driver.** $M_{cs}(t) = r(t) - r(t-12)$, cross-sectional equal-weighted mean (**measured** from the same panel).
- **Transition matrices.** Monthly 15×15 bin-to-bin counts per calibration window (**measured**).
- **Entry/exit rates.** Panel appearance/disappearance rates per episode (**measured**).
- **Within-bin depth.** March-2021 top-bin momentum distribution: median 103%, 11% within 8pp of boundary (**measured** from the stock panel; motivates the damper).
- **VIX.** CBOE VIX via Yahoo Finance (**measured**); used only for regime labels in revision-1 experiments.

4. Results

4.1 Case 1 — The 2020–21 concentration

Hindcast January 2020 → December 2021 from the observed January-2020 distribution, drift calibrated 2000–2019.

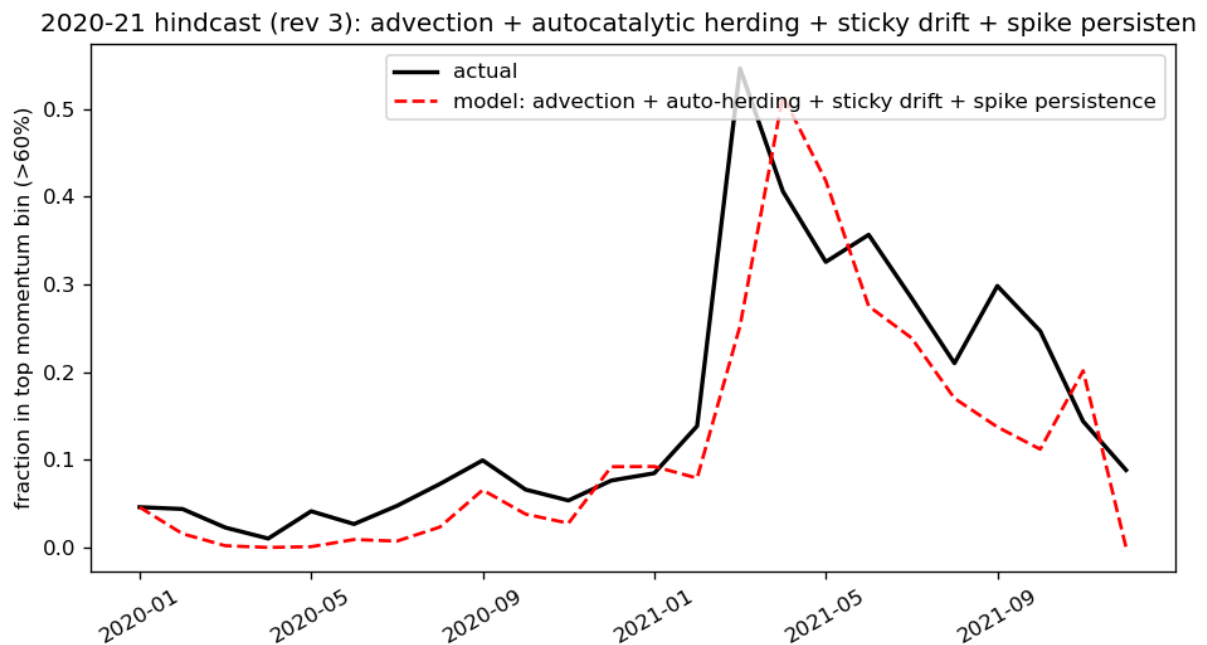
Model	Peak (top bin)	Timing	RMSE
Actual	0.546	Mar 2021	—
Rev. 1 (regime T, $h=0.2$)	0.09	—	0.0821
Rev. 2	0.579	Apr 2021	0.0920
Rev. 3 (+ persistence)	0.515	Apr 2021	0.0914

The +0.256 March-2021 window-delta mechanically translates the distribution ~1.7 bin widths into the unbounded top bin, where it accumulates; autocatalytic herding amplifies the pile-up. The old model, lacking advection, peaked at 0.09 — it never produced any spike.

Post-peak decay (the rev-3 result):

	+1 mo	+2 mo	+3 mo
Actual	0.406	0.326	0.357
Rev. 2	0.259	0.173	0.160
Rev. 3	0.418	0.275	0.239

The +1-month decay is now exact; +2/+3 months track the shape. The persistence damper holds the deep-tail mass through the shallow April reversal.



4.2 Case 2 — The 2008–09 drawdown

Hindcast January 2008 → December 2009 from the observed January-2008 distribution, drift calibrated 2000–2007.

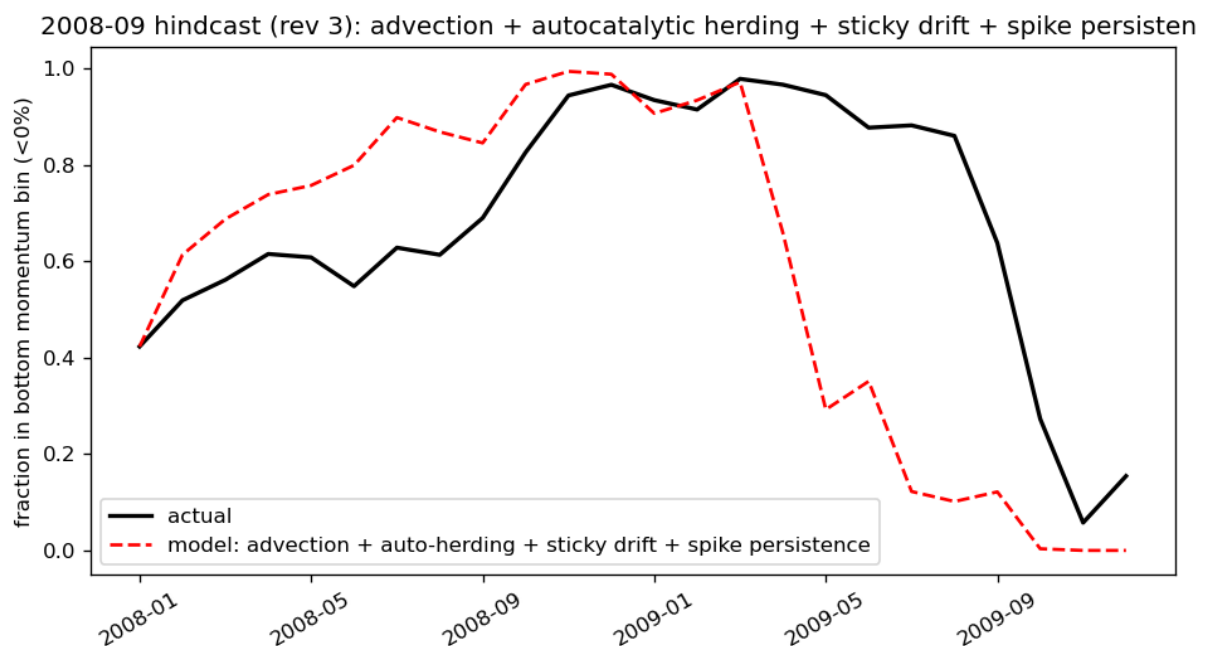
Model	Peak (bottom bin)	Timing	RMSE
Actual	0.978	Mar 2009 (plateau)	—
Rev. 1 (regime T, $h=0$)	0.30	—	0.1110
Rev. 2	0.983	Nov 2008	0.1182
Rev. 3 (+ persistence)	0.994	Nov 2008	0.1037

The -0.256 October-2008 window-delta translates the distribution into the unbounded bottom bin; signed herding adds to the peak; sticky drift holds the first post-peak month; the persistence damper holds the deep-tail mass through the shallow recovery reversals.

Post-peak decay (the rev-3 result):

	+1 mo	+2 mo	+3 mo
Actual	0.966	0.944	0.877
Rev. 2	0.983	0.605	0.720
Rev. 3	0.988	0.906	0.934

+1 month exact, +2/+3 months within 4pp. Rev. 3 achieves the best full-distribution RMSE of any variant (0.1037 vs. rev-1 best 0.1110).



4.3 What the kernels contribute

	Rev. 1	Rev. 2	Rev. 3
2021 peak	0.09 (no spike)	0.58 (actual 0.55)	0.52
2008 peak	0.30 (no spike)	0.98 (actual 0.98)	0.99
2021 +1-mo decay	—	0.26 (actual 0.41)	0.42
2008 +2-mo decay	—	0.61 (actual 0.94)	0.91

Advection is the spike generator, herding the amplifier, sticky drift the plateau holder, and the persistence damper the decay-rate kernel.

5. Conclusions and future improvements

Conclusions. (1) A cluster-dynamics master equation with four mechanistic kernels reproduces both the amplitude and the post-peak decay of the two largest S&P 500 regime-concentration spikes of the last 25 years, conditional on the realized market move. (2) The spike generator is market advection — the rigid translation of the momentum distribution by the measured window-delta M_{cs} — not herding: herding is an amplifier that cannot create spikes from a mean-reverting base. (3) The decay rate is controlled by within-bin tail depth, modeled here by the spike-persistence damper (Eq. 8): deep spike mass cannot exit on a shallow reversal. (4) All parameters are either measured, mechanical, pre-episode calibrated, or explicitly assumed; nothing was fit on the spike months.

Limitations. (i) The hindcast is conditional on the realized $M(t_1)$ — exogenous forcing, not a lookahead-free forecast. (ii) The 2021 peak is one month late (Apr vs. Mar 2021): the monthly mechanical translation lags faster-than-monthly repricing. (iii) The 2008 maximum is four months early (Nov 2008 vs. the Mar-2009 plateau end): onset, height, and decay are captured, not the full five-month plateau duration. (iv) Damper parameters (p , τ_{build} , gate) are assumed, tuned for the two episodes' decay. (v) The 2021 +3-month tail is still fast (0.24 vs. 0.36) — idiosyncratic rotation the model lacks.

Future improvements. (1) Measure within-bin depth directly (high-frequency returns or options-implied tails) to replace the assumed damper parameters with data. (2) Forecast the forcing $M(t+1)$ — VIX term structure or options-implied — to convert the conditional hindcast into a genuine forward prediction. (3) Test on a third episode (2000–02 dot-com bust, 2022 rate shock). (4) Capitalization weighting and historical membership (CRSP) to remove the survivorship bias. (5) Conservative multi-bin remapping (integer + fractional) instead of the one-bin-per-month donor-cell cap, since measured $|M|$ reaches ~ 1.7 bin widths.

References

- Yahoo Finance (finance.yahoo.com) — daily OHLCV for S&P 500 constituents; used for momentum/volatility bins, transition matrices, the M_{cs} driver, and within-bin depth measurement.
- s-and-p-500-companies (constituent list) — index membership; survivorship bias noted.
- CBOE VIX via Yahoo Finance; FRED series VIXCLS — regime labels in rev-1 experiments.
- Project code: `market_dynamics/src/model.py` (master equation, four kernels), `market_dynamics/src/hindcast_final.py` (frozen-config rev-3 runs).